Abdelrahman Eldesokey, PhD

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I am a Postdoctoral Fellow at KAUST, Saudi Arabia, specializing in Generative AI. I work with a team of PhD students conducting research on advanced topics in generative models and Large Language Models (LLMs). I bring over 11 years of combined academic and industrial experience in computer vision and machine learning across Sweden, Egypt, and Saudi Arabia, I hold a Ph.D. in Computer Vision and Deep Learning from Linköping University in Sweden.

AREAS OF EXPERTISE

- **Computer Vision & Machine Learning** Deep Learning, Diffusion Models, Multi-Modal Large Language Models (MLLM), AI for Sports, Depth Estimation, Object Detection/Tracking, Uncertainty Estimation for Deep Learning, Image/Video Object Segmentation, Optical Flow
- **Software Management** Agile, Version Control, Cloud (Azure, AWS), Distributed Systems (Redis, IROS)

PERSONAL INFORMATION

- Date of Birth: 06/11/1989
- Citizenship: Swedish, Egyptian
- Residence: KAUST, Saudi Arabia

PROFESSIONAL EXPERIENCE

- Mar 2023 - Present: ***Postdoctoral Fellow*** at KAUST, Saudi Arabia
  - ***3D Layout Control for Text-to-Image Diffusion Models:*** Introduced a novel approach for controlling image generation through a user-provided 3D layout [Project Page] [Paper]
  - ***Consistent Characters Animation Synthesis using Diffusion Model:*** Developed a zero-shot approach for animating characters based on image and motion diffusion model [Project Page] [Paper]
  - ***Diffusion-Based Video Semantic Segmentation:*** Supervised a project for segmenting videos using pre-trained diffusion models under a zero-shot setting [Project Page] [Paper]
  - ***3D Sematic Correspondence using Vision and Large Language Models:*** Supervised a project for matching non-isometric 3D shapes based on vision and language models [Project Page] [Paper]
  - ***Head Avatars Generation with Multi-Modal Conditioning:*** Supervised a project for generating and editing head avatars by conditioning a GAN using a diffusion model [Paper]
  - ***Fine-Grained Image Editing using Diffusion Models:*** Proposed an approach for editing fine parts of images to produce complex image concepts [Coming Out Soon]
  - ***Improving Reasoning of Multi-Modal LLMs through Enhanced Vision Encoders:*** Developing a visual chain-of-thought paradigm for improving the visual reasoning of MLLM [In progress]
  - ***Multi-View Consistent Vision Features:*** Co-developing a vision encoder based on image foundation models that is consistent across different views [In progress]

- Aug 2021 – Feb 2023: ***Senior Machine Learning Engineer*** at Signality, Sweden
  - ***AI-based Algorithms for Sports Analytics:*** Led a team of 3-5 Machine Learning engineers and data scientists to develop algorithms for player detection/tracking and automated events detection (Auto VAR).
  - ***Scrum Master:*** Managed and tracked the progress of the R&D team to meet the deadlines and deliverables in a timely manner.

- Apr 2022 – Feb 2023: ***Adjunct Lecturer*** at Linköping University, Sweden
  - ***Multi-Modal Uncertainty Estimation in Regression Tasks:*** Supervised a project for estimating uncertainty in regression tasks with multi-modal predictions [Paper]

- May 2016 – Jul 2021: ***PhD Student*** at Linköping University, Sweden.
  - ***Uncertainty Estimation for Sparse Data:*** Developed a novel Convolutional Neural Network (CNN) architecture for uncertainty estimation in sparse data denoted as *Normalized CNNs* with several applications in autonomous driving, structure-from-motion, and robotics.
  - ***Vehicles Detection in Thermal Imagery:*** A collaboration project with the Swedish Defense Research Agency (FOI) where we developed a GAN-based approach for generating synthetic datasets to train object detection models in the thermal domain.
  - ***Cyclist Detection in Driving Environments:*** An industrial/academic collaboration where I developed vision-based cyclist detection algorithms in challenging driving environments.

- Oct 2013 – Apr 2016: ***Research Assistant*** at Nile University, Egypt.
  - ***TraffiSense:*** Developed a smart vision-based traffic analytics system for the Ministry of Interior in Egypt. This included vehicle count, velocity estimation, density analysis, violation detection, and License Plate Recognition (LPR).

EDUCATION

- Nov 2016 – July 2021: ***PhD in Computer Vision and Deep Learning*** at Linköping University, Sweden
  - ***Thesis:*** "Uncertainty-Aware Convolutional Neural Networks for Vision Tasks on Sparse Data"
  - ***Supervisors:*** Prof. Michael Felsberg & Prof. Fahad Shahbaz Khan

- Oct 2013 – Apr 2016: ***MSc in Communication and Information Technology*** at Nile University, Egypt
  - ***Thesis:*** "The AERIAL tracker: A Robust Visual Tracker for Micro-Drones"
  - ***cGPA:*** 3.97/4.0
  - ***Supervisor:*** Prof. Mohamed Elhelw

- Sep 2007– Aug 2011: ***BSc in Computers and Systems Engineering*** at Mansoura University, Egypt
  - ***Graduation Project:*** "Augmented Reality Platform on Android for Educational Applications"
  - ***Grade:*** Excellent with honors (Rank 4/180)

SELECTED PUBLICATIONS

- [2024] **Eldesokey, A.**, & Wonka, P. "Build-A-Scene: Interactive 3D Layout Control for Diffusion-Based Image Generation". *arXiv preprint arXiv:2408.14819*.
- [2024] Wang, Q., **Eldesokey, A.**, Mendiratta, M., Zhan, F., Kortylewski, A., Theobalt, C., & Wonka, P. "Zero-Shot Video Semantic Segmentation based on Pre-Trained Diffusion Models". *arXiv preprint arXiv:2405.16947*.
- [2024] Para, W. R., **Eldesokey, A.**, Li, Z., Reddy, P., Deng, J., & Wonka, P. "AvatarMMC: 3D Head Avatar Generation and Editing with Multi-Modal Conditioning". *arXiv preprint arXiv:2402.05803*.
- [2024] **Eldesokey, A.**, & Wonka, P. "LATENTMAN: Generating Consistent Animated Characters using Image Diffusion Models". In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 7510-7519).
- [2024] Xiong, Z., Jonnarth, A., **Eldesokey, A.**, Johnander, J., Wandt, B., & Forssén, P. E. "Hinge-Wasserstein: Estimating Multimodal Aleatoric Uncertainty in Regression Tasks". In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 3471-3480).
- [2023] Abdelreheem, A., **Eldesokey, A.**, Ovsjanikov, M., & Wonka, P. "Zero-shot 3d shape correspondence". In *SIGGRAPH Asia 2023 Conference Papers* (pp. 1-11).
- [2021] **Eldesokey, A.**, & Felsberg, M. "Normalized Convolution Upsampling for Refined Optical Flow Estimation". In 16th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP 2021), (Vol. 5, pp. 742-752).

- [2021] **Eldesokey, A.** "*Uncertainty-Aware Convolutional Neural Networks for Vision Tasks on Sparse Data"* (Doctoral dissertation, Linköping University Electronic Press).
- [2020] **Eldesokey, A.**, Felsberg, M., Holmquist, K., & Persson, M. "Uncertainty-aware CNNs for depth completion: Uncertainty from beginning to end". In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 12014-12023).
- [2019] **Eldesokey, A.**, Felsberg, M., & Khan, F. S. "Confidence propagation through CNNs for guided sparse depth regression". *IEEE transactions on pattern analysis and machine intelligence*, *42*(10), 2423-2436.
- [2018] Nyberg, A., **Eldesokey, A.**, Bergstrom, D., & Gustafsson, D. "Unpaired thermal to visible spectrum transfer using adversarial training". In *Proceedings of the European conference on computer vision (ECCV) Workshops*.

AWARDS AND HONORS

- Best Paper Award at VISAPP conference (2021)
- Received an Affiliation to the Wallenberg AI, Autonomous Systems and Software Program (2017)
- Graduate Scholarship from Nile University for master's degree (2013)
- Honor award for outstanding performance in BSc from Mansoura University (2011)
- One of the Top 10 BSc graduation projects in Egypt by IEEE Gold (2011)

TECHNICAL SKILLS

- ***Programming***

Language	Years of Experience	Last Used
Python	10 Years	Present
Matlab	10 Years	2019
C/C++	7 Years	2016

- ***Deep Learning Frameworks*** PyTorch (Expert), TensorFlow & Keras (Intermediate)
- ***Data Science*** Pandas, Scikit-learn, Seaborn
- ***Cloud*** AWS, Azure
- ***Versioning Control*** Git
- ***Distributed System*** Redis, IROS
- ***Agile Management***
  - Jira/Confluence
  - Trello

SOFT SKILLS

- ***Languages***
  - Arabic (Mother Tongue)
  - English (Fluent)
  - Swedish (Advanced)
- ***Self-motivated:*** Can steer projects from the planning phase to delivery autonomously
- ***Team Leading:*** Can lead small teams of engineers/students and exploit their individual potential
- ***Presentation Skills:*** Poses excellent presentation skills and can adapt concepts to fit audiences of different backgrounds
- ***Social:*** Can effortlessly interact with people from different backgrounds and get them engaged